

Ansh Maru

+1 (226) 507-2674 | amaru@uwaterloo.ca | anshmaru.pages.dev | linkedin.com/in/anshmaru | github.com/anshm42

EDUCATION

University of Waterloo

Waterloo, Ontario

Bachelor of Mathematics in Computational Mathematics - Honours

Expected Graduation: 2029

- **Relevant Coursework:** Data Structures and Algorithmic Problem Solving, Object Oriented Software Development, Functional Programming, Computer Systems

EXPERIENCE

Patents & Innovation Intern

Sept 2025 – Dec 2025

Pratt & Whitney Canada

Mississauga, Ontario

- Performed open-domain research into **electronic engine control systems** supporting future projects
- Collaborated with engine controls team researching **engine control software and hardware security**
- Deployed an internal PowerBI dashboard used by a cross-functional team of over **30+** members

PROJECTS

Nuclear Leak Probability and Classification | *Python, XGBoost, scikit-learn, pandas, NumPy, Streamlit, joblib*

- **Winner for Best Final Product at the 2026 Canadian Nuclear Labs Nuclear Innovation Challenge**
- Developed calibrated **XGBoost** models to detect reactor leak events and classify the likely leak type from Nuclear Power Plant Accident Data (NPPAD) simulated nuclear-plant sensor time series.
- Prevented train/test contamination by splitting at the complete-simulation level and excluding direct leak-flow variables that would reveal the answer.
- Achieved **100% accuracy identifying probable leaks** and **100% accuracy classifying target leak scenarios** on the simulation test set

ExSAMine | *NextJS, FastAPI, Gemini API, Modal*

- **Runner up for Best Digital Forensics Hack** at HoyaHacks 2026
- Developed AI-powered digital forensics platform using **Next.js** and **FastAPI**, integrating Google Gemini and **Meta's SAM3** for automated evidence analysis
- Built 3-stage AI pipeline: LLM-generated detection prompts to computer vision segmentation to forensic insights generation with confidence scoring
- Deployed object detection model to **Modal serverless GPUs** with automatic scaling and persistent model caching

slangify | *Python, Gemini API, Urban Dictionary API, Docker*

- Built and deployed an always-on Discord bot for explaining slang, regional dialect, jargon, and non-standard language; supports **30+ users** and **70+ lookups** per day.
- Added selectable Gemini-only and Urban Dictionary-only modes, Discord embed formatting, and automated tests for parsing, formatting, and tokenization.
- Containerized and deployed to **Azure Cloud Services** using **Docker**.
- Reduced cloud hosting cost from roughly \$25/month to **\$5/month (80%)** by migrating and right-sizing the Azure deployment.

Neural Network | *C++*

- Engineered a modular multilayer perceptron framework by implementing custom layer and activation interfaces in **C++**, resulting in flexible support for arbitrary network depth.
- Assembled configurable activation modules (ReLU, Leaky ReLU or Sigmoid) selectable per layer to optimize gradient flow and convergence.
- Verified performance on the MNIST dataset, trained via mini-batch gradient descent with backpropagation, and achieving **95%** accuracy when tested.

TECHNICAL SKILLS

Languages: C++, C, C#, Typescript, Javascript, SQL, Python, HTML/CSS

Frameworks: scikit-learn, XGBoost, pandas, Node.js, Flask, FastAPI, React, NextJS, Electron

Tools: Git, Github, Valgrind, gdb, Make, Vim, VS Code, Bash, Zsh, Linux, SQLite, PostgreSQL, L^AT_EX